Baseline Maths Assessment

Year 6

This resource corresponds with the New National Curriculum objectives for Year 5, and has been designed to be used as an aide in assessing pupils' baseline mathematical knowledge at the start of Year 6.

The content addresses each curricular objective outlined in the New National Curriculum in chronological order to give insight into strengths or gaps in each child's mathematical knowledge.

More Assessment resources.

Baseline Maths Assessment – Year 6

Year 5 National Curriculum Objectives:

Number – number and place value

1, 2, 3 & 4. Read, write, order and compare numbers to at least 1 000 000 and determine the value of each digit

5. Count forwards or backwards in steps of powers of 10 for any given number up to 1 000 000

6. Interpret negative numbers in context, count forwards and backwards with positive and negative whole numbers, including through zero

7. Round any number up to 1 000 000 to the nearest 10, 100, 1000, 10 000 and 100 000

9. Solve number problems and practical problems that involve all of the above

8. Read Roman numerals to 1000 (M) and recognise years written in Roman numerals

Number – addition and subtraction

10, 11, 12 & 13. Add and subtract whole numbers with more than 4 digits, including using formal written methods (columnar addition and subtraction)

15, 16, 17 & 18. Add and subtract numbers mentally with increasingly large numbers

14. Use rounding to check answers to calculations and determine, in the context of a problem, levels of accuracy

19. Solve addition and subtraction multi-step problems in contexts, deciding which operations and methods to use and why

Baseline Maths Assessment – Year 6

Year 5 National Curriculum Objectives:

Number – multiplication and division

20 &21. Identify multiples and factors, including finding all factor pairs of a number, and common factors of two numbers

22. Know and use the vocabulary of prime numbers, prime factors and composite (non-prime) numbers

22. Establish whether a number up to 100 is prime and recall prime numbers up to 19

23 & 24. Multiply numbers up to 4 digits by a one- or two-digit number using a formal written method, including long multiplication for two-digit numbers

25. Multiply and divide numbers mentally drawing upon known facts

26. Divide numbers up to 4 digits by a one-digit number using the formal written method of short division and interpret remainders appropriately for the context

27. Multiply and divide whole numbers and those involving decimals by 10, 100 and 1000

28. Recognise and use square numbers and cube numbers, and the notation for squared (2) and cubed (3)

29. Solve problems involving multiplication and division including using their knowledge of factors and multiples, squares and cubes

30. Solve problems involving addition, subtraction, multiplication and division and a combination of these, including understanding the meaning of the equals sign

31.Solve problems involving multiplication and division, including scaling by simple fractions and problems involving simple rates

Baseline Maths Assessment – Year 6

Year 5 National Curriculum Objectives:

Number – fractions (including decimals and percentages)

32. Compare and order fractions whose denominators are all multiples of the same number

33. Identify, name and write equivalent fractions of a given fraction, represented visually, including tenths and hundredths

34. Recognise mixed numbers and improper fractions and convert from one form to the other and write mathematical statements > 1 as a mixed number [for example, 2/5 + 4/5 = 6/5 = 1 1/5]

35. Add and subtract fractions with the same denominator and denominators that are multiples of the same number

36. Multiply proper fractions and mixed numbers by whole numbers, supported by materials and diagrams

37. Read and write decimal numbers as fractions [for example, 0.71 = 71/100]

38. Recognise and use thousandths and relate them to tenths, hundredths and decimal equivalents

39. Round decimals with two decimal places to the nearest whole number and to one decimal place

40. Read, write, order and compare numbers with up to three decimal places

41.Solve problems involving number up to three decimal places

42. Recognise the per cent symbol (%) and understand that per cent relates to 'number of parts per hundred', and write percentages as a fraction with denominator 100, and as a decimal

43. Solve problems which require knowing percentage and decimal equivalents of 1/2 , 1/4 , 1/5 , 2/5 , 4/5 and those fractions with a denominator of a multiple of 10 or 25

Baseline Maths Assessment – Year 6

Year 5 National Curriculum Objectives:

Measurement

44. Convert between different units of metric measure (for example, kilometre and metre; centimetre and metre; centimetre and millimetre; gram and kilogram; litre and millilitre)

45. Understand and use approximate equivalences between metric units and common imperial units such as inches, pounds and pints

46. Measure and calculate the perimeter of composite rectilinear shapes in centimetres and metres

47. Calculate and compare the area of rectangles (including squares), and including using standard units, square centimetres (cm2) and square metres (m2) and estimate the area of irregular shapes

48. Estimate volume [for example, using 1 cm3 blocks to build cuboids (including cubes)] and capacity [for example, using water]

49. Solve problems involving converting between units of time

50. Use all four operations to solve problems involving measure [for example, length, mass, volume, money] using decimal notation, including scaling

Geometry – properties of shapes

51.Identify 3-D shapes, including cubes and other cuboids, from 2-D representations

52. Know angles are measured in degrees: estimate and compare acute, obtuse and reflex angles

* Draw given angles, and measure them in degrees

* Identify angles at a point and one whole turn (total 360)

* Identify angles at a point on a straight line and 1/2 a turn (total 180)

* Identify other multiples of 90

53 & 54. Use the properties of rectangles to deduce related facts and find missing lengths and angles

55. Distinguish between regular and irregular polygons based on reasoning about equal sides and angles

*these objectives are practical so are not included in this assessment

Baseline Maths Assessment – Year 6

Year 5 National Curriculum Objectives:

Geometry – position and direction

56 & 57. Identify, describe and represent the position of a shape following a reflection or translation, using the appropriate language, and know that the shape has not changed

Statistics

59. Solve comparison, sum and difference problems using information presented in a line graph

58 & 60. Complete, read and interpret information in tables, including timetables

Baseline Maths Assessment - Year 6

Number and Place Value

1. Write the following numbers in words.

104,789 _____

850,045 _____

2. Write the follow numbers in digits.

Three hundred and six thousand, two hundred and one. _____

One million, nine hundred-thousand, seven hundred and eighty. _____

3. Use the symbols > or < to compare these numbers.

124,509 ☐ 124,590 506,476 ☐ 506,467 259,000 ☐ 250,900

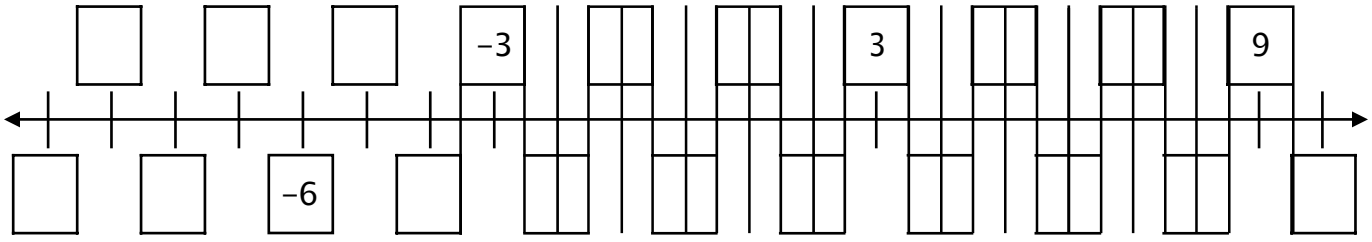
4. Order the following numbers from smallest to largest.

345,560	354,560	309,560	390,560	390,506
☐	☐	☐	☐	☐

5. Continue the sequence of numbers.

105 115 125 135 ☐ ☐ ☐ ☐

6. Fill in the missing numbers on the number line.



7. Round the following numbers to the nearest 10.

67 _____ 89 _____ 54 _____

Round the following numbers to the nearest 100.

456 _____ 739 _____ 949 _____

Round the following numbers to the nearest 1000.

3501 _____ 6099 _____ 8604 _____

Round the following numbers to the nearest 10,000.

14,569 _____ 25,690 _____ 99,009 _____

8. Write down the number that these Roman numerals represent.

XX _____ XC _____

XL _____ D _____

LXX _____ M _____

9. A pupil has forgotten the 4 digit code to her locker but luckily, she wrote down clues to remind her. The clues are as follows:

- The third number is twice the value of the first number.
- The second number is a prime number greater than 5 but less than 11.
- The first number is greater than 1, an odd number and a factor of 6.
- The fourth number is the product of 3 multiplied by itself.

The code is: _____ _____ _____ _____

Addition and Subtraction

10. 4563 + 3956 =

11. 56,112 + 30,462 =

12. 82,908 – 41,572 =

13. 92,231 – 73,999 =

14. Circle the incorrect calculation.

341 + 679 = 1020 783 – 291 = 492 1292 + 459 = 1731 879 – 341 = 538

15. 458 + ☐ = 670

16. ☐ + 390 = 798

17. ☐ − 465 = 358

18. 942 − ☐ = 330

19. 164 passengers got on an empty train in Birmingham. 198 more passengers got on at Wolverhampton, but 76 passengers got off.
How many passengers were on the train as it left the station to continue its onward journey?

Multiplication and Division

20. Find the next 6 multiples of:

12 ☐ ☐ ☐ ☐ ☐ ☐

15 ☐ ☐ ☐ ☐ ☐ ☐

21. Find the common factors of the following 2 numbers.

12 and 16 ☐ ☐ ☐

22. Circle all of the prime numbers in this list.

2 9 11 13 15 17 19 21 23

23. 4563 x 9 =

24. 3217 x 23 =

25.

60 x 70 = ☐ 560 ÷ 7 = ☐ 12 x 300 = ☐

26. 4144 ÷ 7 =

27. Multiply or divide the following numbers by 10, 100 or 1000.

167	x	10	=		1.45	x	100	=	
2390	÷	10	=		4568	÷	100	=	
1.23	x	1000	=		7809	÷	1000	=	

28. Calculate the following squared and cubed numbers.

$4^2 =$ ☐ $9^2 =$ ☐ $5^3 =$ ☐ $7^3 =$ ☐

29. I am a square number.
I am less than 60.
I am a multiple of 7.
What number am I?

30. Sally drives a bus. A return journey from the car park to the city centre is 18.5 km. On Monday, Tuesday and Wednesday, she completes the journey twice. How many kilometres does Sally drive in total?

31. Jamal is given £6 each time he washes the car, and £9 each time he helps with gardening. So far, he has washed the car 18 times and helped with the gardening 6 times. He is given £125. How much more should he be paid?

Fractions (including decimals and percentages)

32. Order the following fractions from smallest to largest.

$\frac{2}{6}$ $\frac{6}{8}$ $\frac{1}{4}$ $\frac{2}{3}$ $\frac{2}{4}$

33. Complete these equivalent fractions.

$\frac{3}{4} = \frac{\quad}{12}$ $\frac{3}{6} = \frac{\quad}{12}$ $\frac{3}{5} = \frac{\quad}{10}$ $\frac{6}{8} = \frac{\quad}{12}$ $\frac{2}{4} = \frac{\quad}{6}$

34. Convert these improper fractions to mixed numbers.

$\frac{9}{6} = \square$ $\frac{14}{8} = \square$ $\frac{12}{5} = \square$ $\frac{16}{7} = \square$ $\frac{31}{9} = \square$

35. Complete the following calculations.

$\frac{1}{3} + \frac{1}{6} = \square$ $\frac{1}{3} + \frac{1}{5} = \square$ $\frac{1}{2} - \frac{1}{6} = \square$

36. Multiply the following fractions.

$\frac{1}{2} \times 5 = \square$ $\frac{1}{4} \times 6 = \square$ $\frac{2}{3} \times 5 = \square$

37. Write the decimals in the correct place on the table.

Less than $\frac{1}{2}$	Equal to $\frac{1}{2}$	More than $\frac{1}{2}$

0.12 0.81
0.1 0.55
0.5 0.72
0.45 0.07

38. Write these fractions as decimal numbers.

$\frac{1}{10}$		$\frac{1}{100}$		$\frac{1}{1000}$	
$\frac{8}{10}$		$\frac{8}{100}$		$\frac{8}{1000}$	

39. Round each amount to the nearest pound.

£2.45		£3.64		£5.99		£4.55	

40. Order these numbers from smallest to largest.

5.032	3.053	5.103	3.335	3.055

41. The Chester family are off on holiday to Cornwall which is 468 km away. They travel 240.78 km before stopping for a break. They then drive a further 152.64 km before getting a flat tyre. How much further do they need to travel in order to reach their destination?

42. Draw a line to match the fraction to the equivalent percentage.

$\frac{50}{100}$	75%
$\frac{25}{100}$	45%
$\frac{75}{100}$	25%
$\frac{45}{100}$	50%

43. Which is more: 50% of 290, or $\frac{1}{4}$ of 224. Explain your reasoning.

Measurement

44. Convert the following litres to millilitres.

2 l	ml	3½ l	ml	5¼ l	ml

Convert the following metres to centimetres.

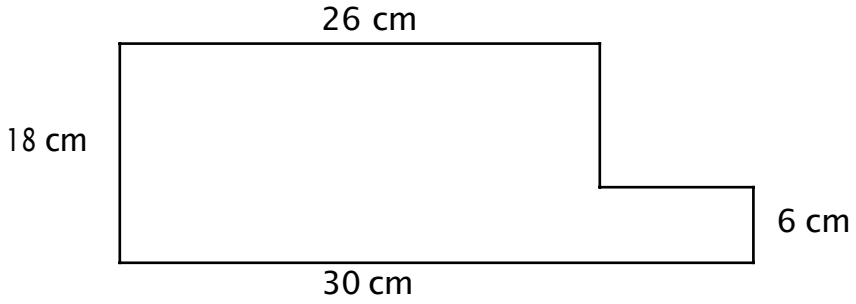
1 m	cm	1.3 m	cm	½ m	cm

45. The world's tallest man measures 251 centimetres. Use the conversion chart below to work out approximately how tall he is in feet and inches.

Conversions

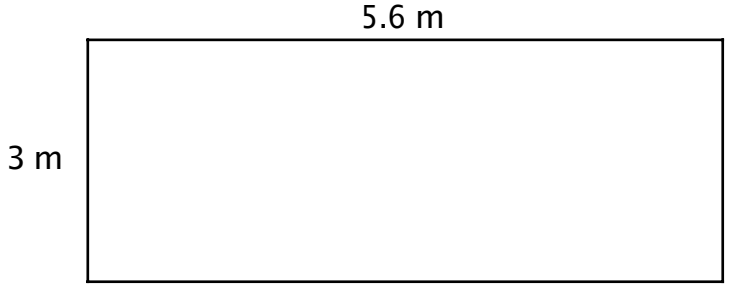
5.08 cm = 2 inch
12 inches = 1 foot

46. Calculate the perimeter of the following shape.



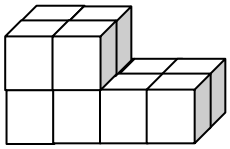
Perimeter = _____cm

47. Calculate the area of the shape.



Area = _____m^2

48. Calculate the volume of the shape.



Volume = _____ cm^3

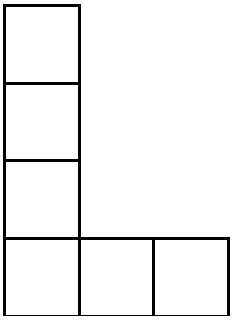
49. A film started at 6.45 pm and finished at 8.25 pm. How long was the film in minutes?

50. A garden measures 34 metres by 9 metres. The family want to cover it in artificial grass. The cost of the artificial grass is £3 per square metre. How much will it cost to cover the whole garden?

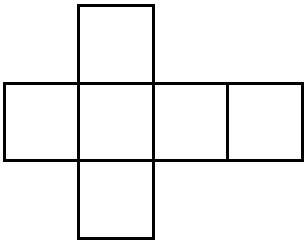
Geometry (properties of shapes)

51. Circle the net that will fold up to create a cube.

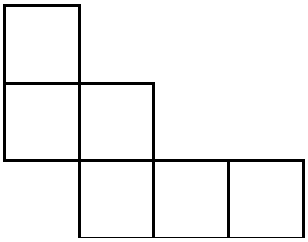
a.



b.

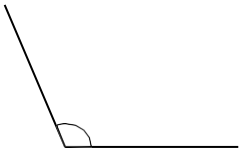


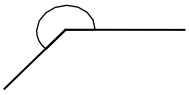
c.

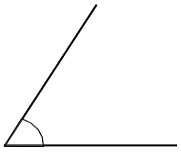


52. Add the correct label the following angles.

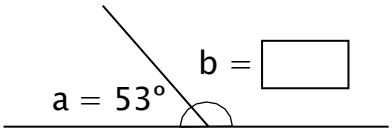
acute obtuse right angle reflex straight line



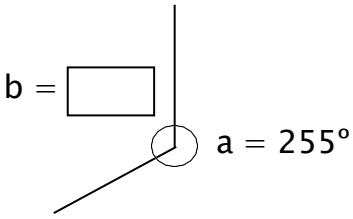




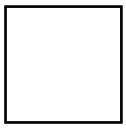
53. Calculate the missing angle.

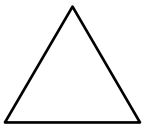


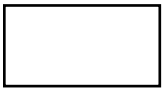
54. Calculate the missing angle.

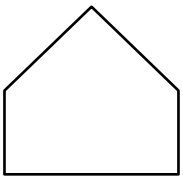


55. Circle the irregular polygons.



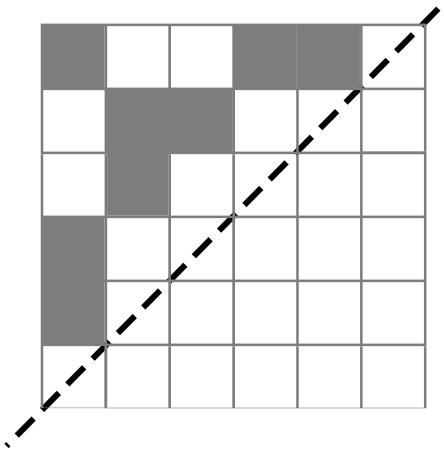


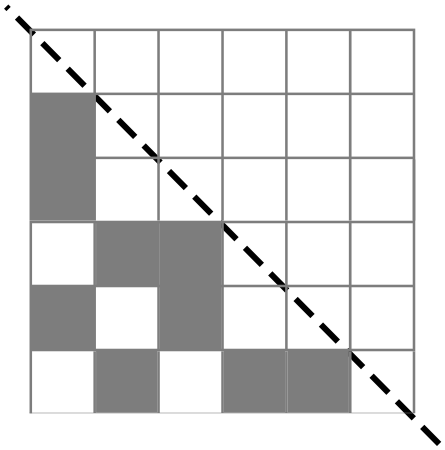




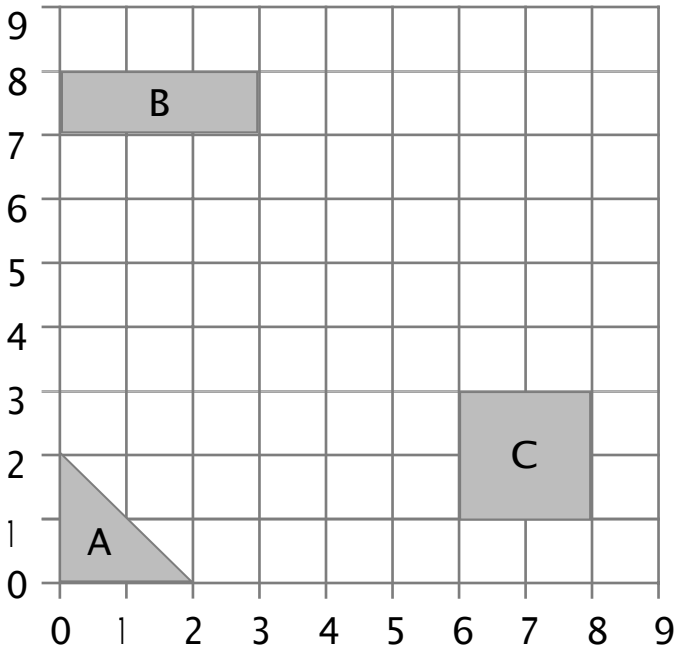
Geometry (position and direction)

56. Shade in the squares to reflect the shapes in the mirror line.





57. Follow the instructions to translate each shape.



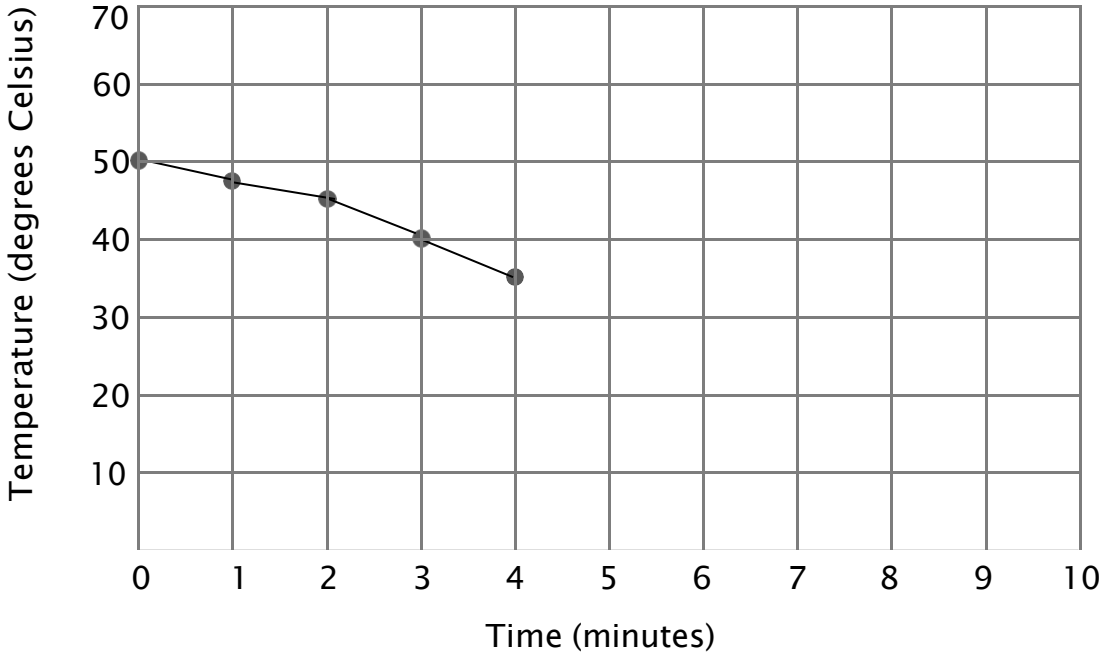
Instructions

- Translate shape A 4 squares up and 2 squares to the right.
- Translate shape B 6 squares to the right and 1 square down.
- Translate shape C 1 square down and 3 squares to the left.

Statistics

58. Complete the line graph using the data below.

Time (mins)	0	1	2	3	4	5	6	7	8	9	10
Temperature (degrees Celsius)	50	48	45	40	35	32	30	27	25	20	20



59. By how many degrees did the temperature decrease between minutes 2 and 3?

How long did the water take to cool down to 27°C?

60. Use the information in the table to answer the following questions.

Destination	Train A	Train B
Stafford	10:25	11:15
Cannock	10:40	____
Walsall	10:52	____
Birmingham	11:30	11:55
Oxford	13:12	13:30

- How long does it take Train A to get from Stafford to Walsall?
- Train A arrives at Oxford 15 minutes early. At what time does it arrive?
- If you travelled from Stafford to Oxford, how much time would you save by taking Train B?

Baseline Maths Assessment – Year 6

Number and Place Value

1. One hundred and four thousand, seven hundred and eighty-nine.
   Eight hundred and fifty thousand, and forty-five.

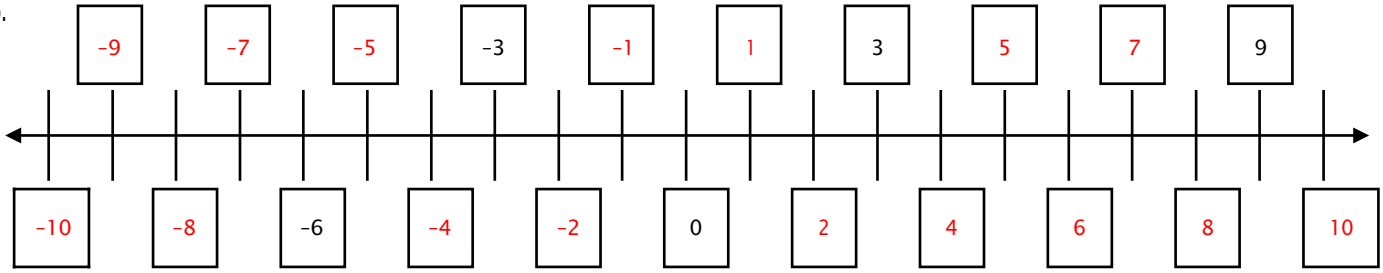
2. 306,201
   1,900,780

3. <, >, >

4. 309,560 345,560 354,560 390,506 390,560

5. 145, 155, 165, 175

6.



7. 70, 90, 50
   500, 700, 900
   4000, 6000, 9000
   10,000, 30,000, 100,000

8.

XX	20	XC	90
XL	40	D	500
LXX	70	M	1000

9. 3769

Addition and Subtraction

10. 8,519

11. 86,574

12. 41,336

13. 18,232

Baseline Maths Assessment – Year 6

14. 1292 + 459 = 1731

15. 212

16. 408

17. 823

18. 612

19. 286

Multiplication and Division

20. 24, 36, 48, 60, 72, 84
    30, 45, 60, 75, 90, 105

21. 1, 2, 4

22. 2, 11, 13, 17, 19, 23

23. 41,067

24. 73,991

25. 4200 80 3600

26. 592

27. 1670 145
    239 45.68
    1230 7.809

28. 16, 81, 125, 343

29. 49

30. 3 x 2 x 18.5 = 111 km

31. £37

Baseline Maths Assessment – Year 6

Fractions (including decimals and percentages)

32. $\frac{1}{4}$ $\frac{2}{6}$ $\frac{2}{4}$ $\frac{2}{3}$ $\frac{6}{8}$

33. $\frac{3}{4} = \frac{9}{12}$ $\frac{3}{6} = \frac{6}{12}$ $\frac{3}{5} = \frac{6}{10}$ $\frac{6}{8} = \frac{9}{12}$ $\frac{2}{4}$ $\frac{3}{6}$

34. $1\frac{3}{6}$ $1\frac{6}{8}$ $2\frac{2}{5}$ $2\frac{2}{7}$ $3\frac{4}{9}$

35. $\frac{3}{6}$ or $\frac{1}{2}$ $\frac{8}{15}$ $\frac{2}{6}$ $\frac{1}{3}$

36. $2\frac{1}{2}$ $1\frac{2}{4}$ or $1\frac{1}{2}$ $3\frac{1}{}$

37.

Less than $\frac{1}{2}$	Equal to $\frac{1}{2}$	More than $\frac{1}{2}$
0.12	0.5	0.81
0.1		0.55
0.45		0.72
0.07		

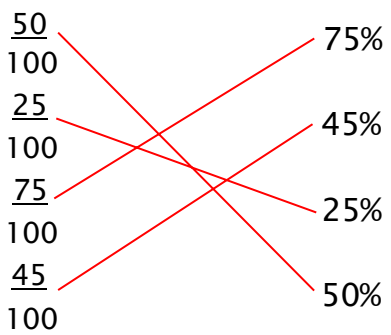
38. 0.1 0.01 0.001
0.8 0.08 0.008

39. £2, £4, £6, £5

40. 3.053, 3.055, 3.335, 5.032, 5.103

41. 74.58 km

42.



43. 50% of 290 = 145, whereas $\frac{1}{4}$ of 224 = 56. So 50% of 290 is more.

Baseline Maths Assessment – Year 6

Measurement

44. 2000 ml, 3500 ml, 5250 ml
    100 cm, 130 cm, 50 cm

45. 8 feet 2 inches / 8 feet 3 inches tall

46. 96 cm

47. 16.8 m^2

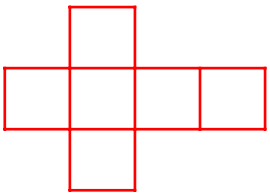
48. 12 cm^3

49. 100 minutes

50. 34 x 9 x £3 = £918

Geometry (properties of shapes)

51. b.

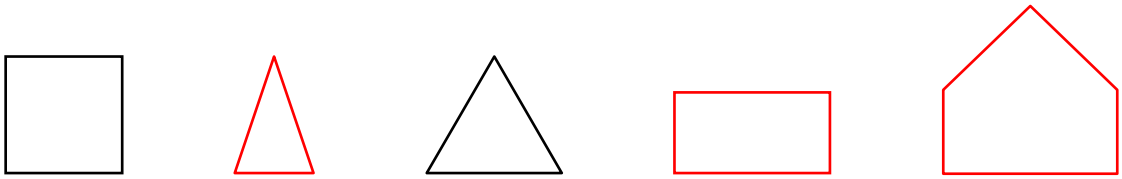


52. obtuse, right angle, reflex, straight line, acute

53. 127°

54. 105°

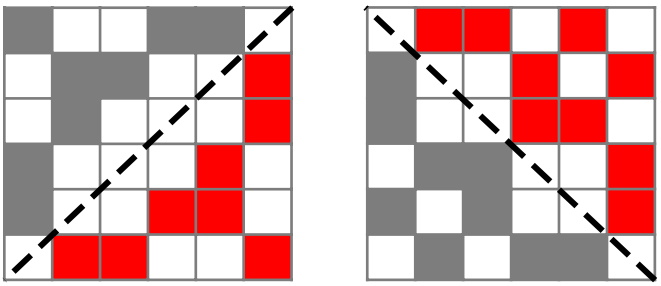
55.



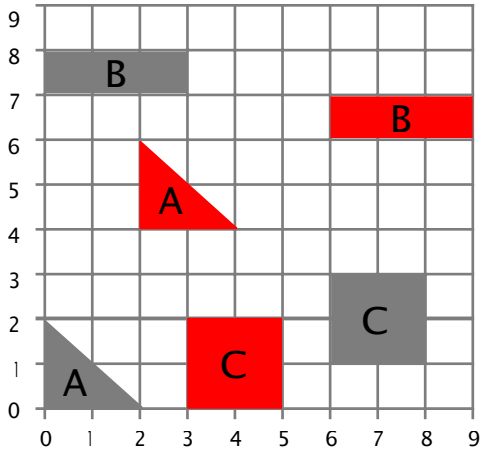
Baseline Maths Assessment – Year 6

Geometry (position and direction)

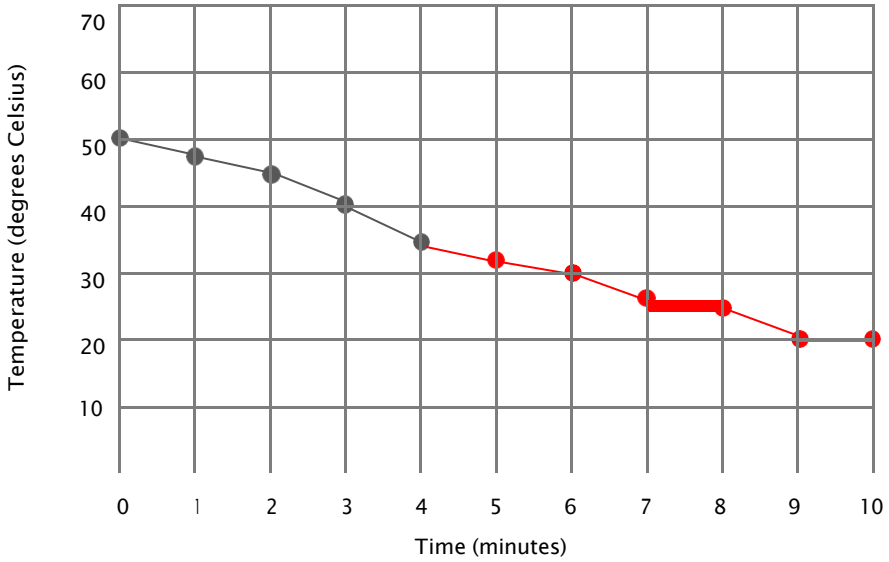
56.



57.



58.



59. 5°C
7 minutes

60. 27 minutes
12:57
32 minutes